DWBA/CRC Calculations for the Multi-Step Investigation of the Reaction ^{25}Mg(d,p) at $E_d = 16$ MeV

K. Hanselman[1]

[1] *Department of Physics, Florida State University, Tallahassee, FL*

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DWBA/CRC calculations for the reaction ^{25}Mg(d,p) complementary to this experiment performed at Florida State University's John D. Fox Linear Accelerator Laboratory (at $E_d = 16$ MeV) are outlined below. First is a verification of the single-step DWBA treatment used by past authors (at 13 MeV beam energy), followed by an extension of the analysis into the multi-step framework of CRC. All steps make use of the code `FRESCO` [1].

I. COMPARISON TO PREVIOUS DWBA TREATMENT

The reaction ^{25}Mg(d,p) was studied at a (lab) deuteron energy of 13 MeV by Arciszewski et al. [2]. They measured the angular distributions of multiple states and assigned spin/parity values and spectroscopic factors to these states. They performed their DWBA calculations using the code `DWUCK`, sans non-locality or finite-range corrections.

Their parameter sets were as follows. For the entrance channel optical potential $^{25}\mathrm{Mg}+d$ they took the parameters of [3]. For the exit channel $^{26}\mathrm{Mg}+p$ they used the scattering results of [4]. The wavefunction of the transferred neutron was their own parameterization, whose depth was varied to match the experimental binding energy. No specification was given for the binding potential of the deuteron. These are summarized in Tab. I.

It is worth noting that the particular entrance channel set chosen by Arc. et al. corresponds in the Perey and Perey tables to $^{24}\mathrm{Mg}+d$ scattering at 11.8 MeV incident (lab) deuteron energy. This particular set was (according to Perey and Perey) a private communcation between G. R. Satchler and P. E. Hodgson in 1965.

The exit channel set from Becchetti et al. was specified for mass ranges $A > 40$ (due to the data fit for both protons and neutrons).

For the DWBA portion of the current analysis, these potentials were deemed sufficient. The parameters where necessary have been converted to the form of the optical potential used by `FRESCO`:

$$V(r) = V_C(r) - V_0 f_0(r) - 4iW_S a_S \frac{d}{dr} f_S(r) - \frac{1}{r} V_{so} \left(\frac{\hbar}{m_\pi c}\right)^2 \frac{d}{dr} f_{so}(r) \boldsymbol{l} \cdot \boldsymbol{\sigma}, \tag{1}$$

where $f_i(r) = 1/[1+\exp(-(r-R_i)/a_i)]$ is the standard Woods-Saxon radial form factor. Here $R_i \equiv r_i \times A_H^{1/3}$, where A_H is the mass number of the "heavier" of the pair. In order these terms are: the Coulomb (charged sphere) potential, the nuclear volume term, the imaginary (absorptive) surface term, and the spin-orbit term.

The only parameterization which had to be specified beyond Arciszewski et al. was that of the deuteron binding potential. This was taken to be the `FRESCO`-standard Reid soft-core potential, which takes into account the (small) d-wave component of the proton-neutron relative motion.

As a first test, these potentials were used for the 16 MeV data, altered appropriately for the slightly different outgoing proton energy in the exit channel. No similar adjustment could be made for the (more important) entrance channel due to the energy dependence of this potential not being given by P&P. Spectroscopic factors were then fit to the data using the fitting counterpart to `FRESCO`, `SFRESCO`, and compared to the 13 MeV results of Arciszewski et al. For the closest comparison, the non-finite-range Local Energy Approximation (LEA) was used, being a standard option in `FRESCO` not requiring any further input. Starting amplitude seed values of both unity and the results of Arciszewski et al. were used, though both yielded identical results.

The fitted 16 MeV spectroscopic factors compared to those of Arciszewski et al. are tabulated in Tab. II. These were performed for the three excited states of astrophysical interest, as well as the first excited state of ^{26}Mg (due to its being used for absolute normalization). Plots of the fitted results against the original spectroscopic factors used by Arciszewski et al. are shown in Fig's 1–4.

Also included in Tab. II are the shell model predictions given by Arciszewski et al. based on their own calculations.

Only results for positive parity states are given, as the calculations were restricted to the $s-d$ shell above an inert ^{16}O core. For the sake of direct comparison, in the SFRESCO fits the same j and l components were assumed as did Arc. et al., regardless of the shell model predictions.

	V_0 (MeV)	r_0 (fm)	a_0 (fm)	W_S (MeV)	r_S (fm)	a_S (fm)	V_{so} (MeV)	r_{so} (fm)	a_{so} (fm)	r_C (fm)
Entr. Channel	93.8	1.00	0.87	29.2	1.36	0.61	12.0	1.00	0.87	1.30
Exit Channel	56.6-0.32E_p	1.17	0.75	12.3-0.25E_p	1.32	0.54	6.2	1.01	0.75	1.37
Neut. Binding	–	1.25	0.65	0.00	0.00	0.00	12.5	1.25	0.65	1.25

TABLE I. Set of specified optical potentials used by [2] for DWBA analysis of the reaction ^{25}Mg(d,p) at $E_d = 13$ MeV. The dash for the neutron binding potential depth indicates that it was varied automatically to match the experimental binding energy for each state.

	Arc. et al. [2]	This Work (LEA)	This Work (FR)	Arc. SM [2]	`CoSMo` [5]
1.806 MeV (2^+)					
$1s_{1/2}$	0.0320(100)	0.0902(232)	0.0707(197)	0.0280	0.0341
$0d_{3/2}$	0.7021(700)	0.5306(746)		0.0100	0.0109
$0d_{5/2}$			0.2343(375)	0.3220	0.3478
Fit χ^2/N		1.41	0.99		
6.125 MeV (3^+)					
$1s_{1/2}$	0.1057(129)	0.1853(167)	0.1743(170)	0.1100	0.1412
$0d_{3/2}$	0.6000(1429)	0.3768(362)	0.2336(235)	0.2957	0.2803
$0d_{5/2}$				0.0343	0.0484
Fit χ^2/N		5.25	4.14		
6.744 MeV (2^+)					
$1s_{1/2}$	0.0080(40)	0.0281(127)	0.0397(136)	0.0060	0.0087
$0d_{3/2}$	0.1160(400)	0.2677(284)		<0.01	0.0995
$0d_{5/2}$			0.1612(165)	0.0840	0.0003
Fit χ^2/N		1.33	1.15		
6.878 MeV (3^-)					
$1p_{1/2}$				–	0.0012
$1p_{3/2}$	0.1686(214)	0.2231(83)	0.2338(95)	–	0.0808
$0f_{5/2}$				–	0.0206
$0f_{7/2}$	0.5000(1429)	0.3345(394)	0.0642(319)	–	0.3512
Fit χ^2/N		3.73	4.04	–	

TABLE II. Fitted spectroscopic (^{25}Mg/^{26}Mg) overlaps compared to those of [2]. Included are shell model predictions performed by Arc. et al. in the $s-d$ shell, and those of a more recent calculation via `CoSMo` [5] in the full $sd-fp$. Each component is denoted by the spin and orbital quantum numbers of the transferred neutron.

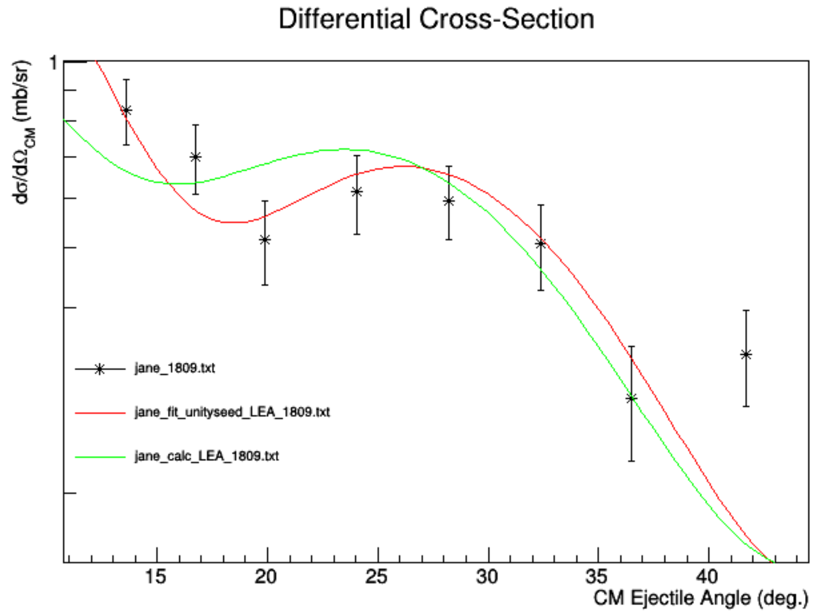

FIG. 1. ^{25}Mg$(d,p)^{25}$Mg(1.809 MeV, 2^+) Comparison of the fitted spectroscopic factors of this work (red) to those of Arciszewski et al. [2] (green). See Tab. II for values. Both calculations use the potentials of Arc. adjusted for 16 MeV. LEA approximation.

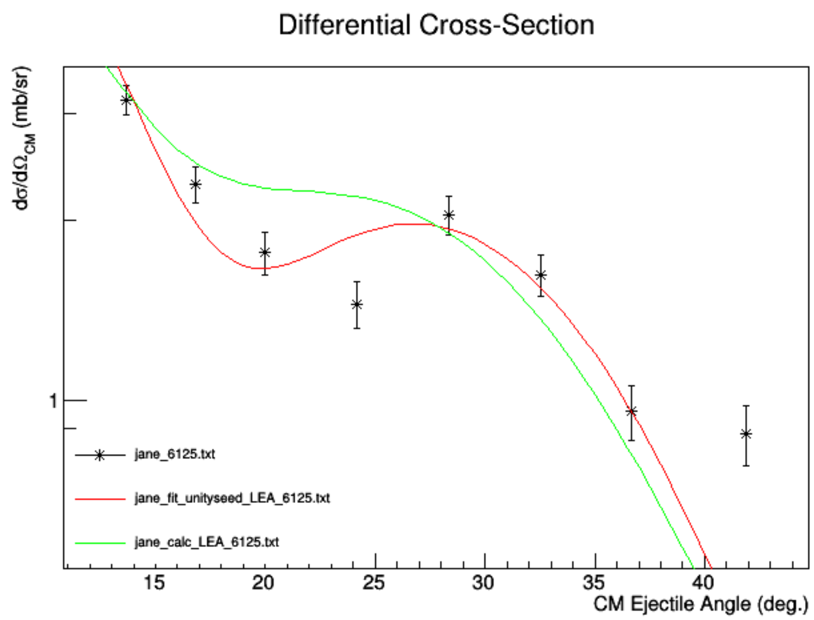

FIG. 2. ^{25}Mg$(d,p)^{25}$Mg(6.125 MeV, 3^+) Comparison of the fitted spectroscopic factors of this work (red) to those of Arciszewski et al. [2] (green). See Tab. II for values. Both calculations use the potentials of Arc. adjusted for 16 MeV. LEA approximation.

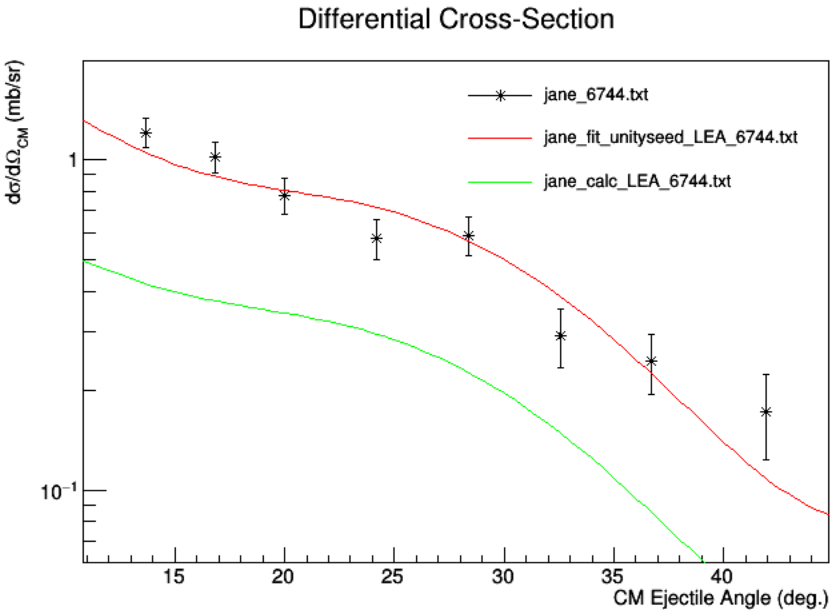

FIG. 3. ^{25}Mg$(d,p)^{25}$Mg(6.744 MeV, 2^+) Comparison of the fitted spectroscopic factors of this work (red) to those of Arciszewski et al. [2] (green). See Tab. II for values. Both calculations use the potentials of Arc. adjusted for 16 MeV. LEA approximation.

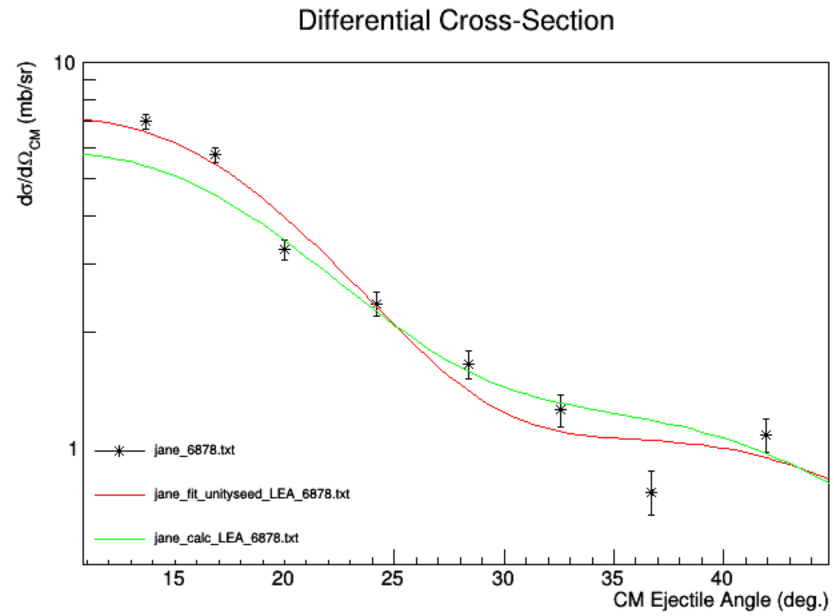

FIG. 4. ^{25}Mg$(d,p)^{25}$Mg(6.878 MeV, 3^-) Comparison of the fitted spectroscopic factors of this work (red) to those of Arciszewski et al. [2] (green). See Tab. II for values. Both calculations use the potentials of Arc. adjusted for 16 MeV. LEA approximation.

II. FINITE-RANGE TREATMENT WITH MODERN OPTICAL POTENTIALS

It is desirable in the context of the current work to take the previous study of Arciszewski et al. [2] and "modernize" it in the following ways: 1) by including full finite-range effects and corrections; and 2) by using more up-to-date global optical potential models for nucleon and deuteron scattering based on more recent experimental data. For the time being, other relevent yet more complicated effects – coupled-channels target excitation, deuteron breakup – will be neglected.

Only data for the (less important) exit channel $^{26}\text{Mg} + p$ exist at this energy range. This scattering was studied by Sciuccati et al. [6] in the (lab) proton energy range from 15–38 MeV, and included fits to inelastic scatterings of excited states up to the present work's region of interest. This energy range covers largely that of the proton's kinetic energy after the present $^{26}\text{Mg}(d, p)$ reaction at $E_d = 16$ MeV.

The optical potential determined by Sciuccati et al. features a linear dependence on the proton kinetic energy in the real and imaginary potential depths (both volume and surface). However, for the range of kinetic energies concerning the present experiment (∼18–23 MeV for forward angles and up to 6 MeV residual ^{26}Mg excitation), this linear correction was found to be minimally varying. Additionally, the exit channel proves to be less impactful on the calculation than that of the entrance channel. Thus, for all angles and excitations, a singular value of proton kinetic energy of 20 MeV was chosen, fixing the exit channel potential parameters to a single set (Tab. III).

For the (more important) entrance $^{25}\text{Mg} + d$ channel, no energy-relevant data are available. Therefore the global deuteron optical model potentials of Haixia and Chonghai [7] were chosen. These potentials were fit from the nuclear range $A = 12$ to $A = 238$, over an energy range of 11.8–183 MeV. For the lighter nuclei in the present region of interest ($A \sim 25$), the energy range was sufficiently on the lower end, and fits the scattering data well at the lower angles. It is also worth noting that this study was an extension of previous studies expanding the original Perey and Perey [3] work used by Arciszewski et al. These are also given in Tab. III.

For the proton-neutron binding potential in the deuteron wave function, the Reid Soft-Core potential has been retained. This is also true of the parameters of the $^{25}\text{Mg} + n$ binding potential, with the exception of the spin-orbit depth V_{so} being reduced to 6 MeV to match more modern conventions. As such, only the new entrance and exit potential parameters are given in Tab. III.

It is important to note that unlike the original channel potential sets used by Arciszewski et al., these both include a volume imaginary absorption component W_V as $-iW_V f_V(r)$. However, at such low particle energies this nuclear volume contribution to the absorption is small, such that the surface effects dominate. For the (global) entrance channel potential this small depth was nevertheless comparable enough to be retained. For the exit channel set a la Sciuccati et al., this imaginary volume depth has been neglected, being less than an MeV in strength.

These potentials have been used in a full finite-range single-step DWBA calculation for the same states. The post form of the interaction was used, including full complex remnant correction terms. This time the choice of j, l was motivated by the shell model predictions, although only one j for a given l was fitted due to the relative insensitivity of even finite-range calculations to the spin over the orbital angular momentum. The resulting fitted spectroscopic factors are displayed in the appropriate column in Tab. II. The plots of these calculations (compared to the original LEA results) are shown in Fig's 5–8.

	V_0 (MeV)	r_0 (fm)	a_0 (fm)	W_V (MeV)	r_V (fm)	a_V (fm)	W_S (MeV)	r_S (fm)	a_S (fm)	V_{so} (MeV)	r_{so} (fm)	a_{so} (fm)	r_C (fm)
Entr. Channel [7]	90.53	1.149	0.756	2.10	1.339	0.563	10.34	1.386	0.712	3.56	0.972	1.011	1.303
Exit Channel [6]	49.25	1.150	0.650	–	–	–	4.48	1.132	0.600	5.60	1.010	0.600	1.200

TABLE III. Set of global optical potentials for the use of finite-range DWBA analysis of the reaction ^{25}Mg(d,p) at $E_d = 16$ MeV.

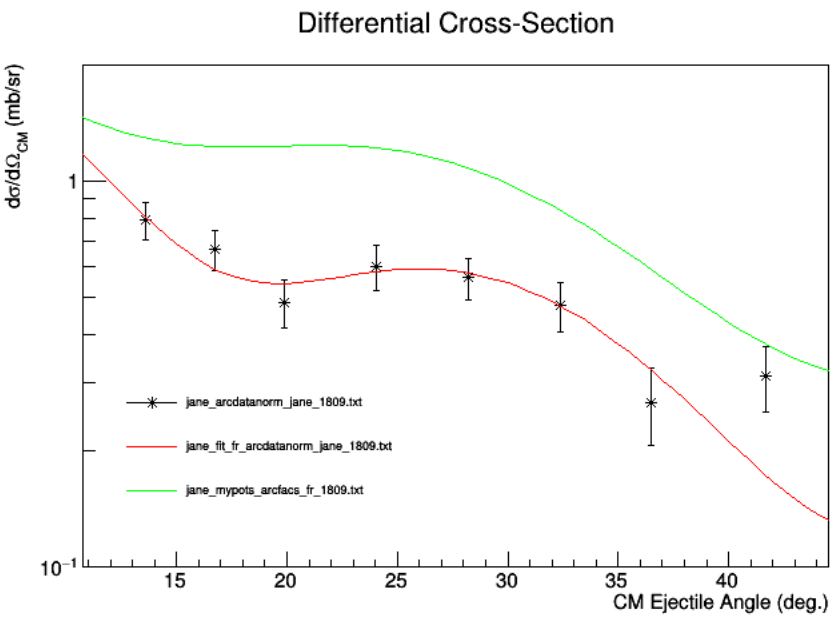

FIG. 5. ^{25}Mg$(d,p)^{25}$Mg(1.809 MeV, 2^+) Comparison of fitted spectroscopic factors for the current work (red) and those of Arc. et al (green). Both use the finite-range approximation and the potentials of Tab. III.

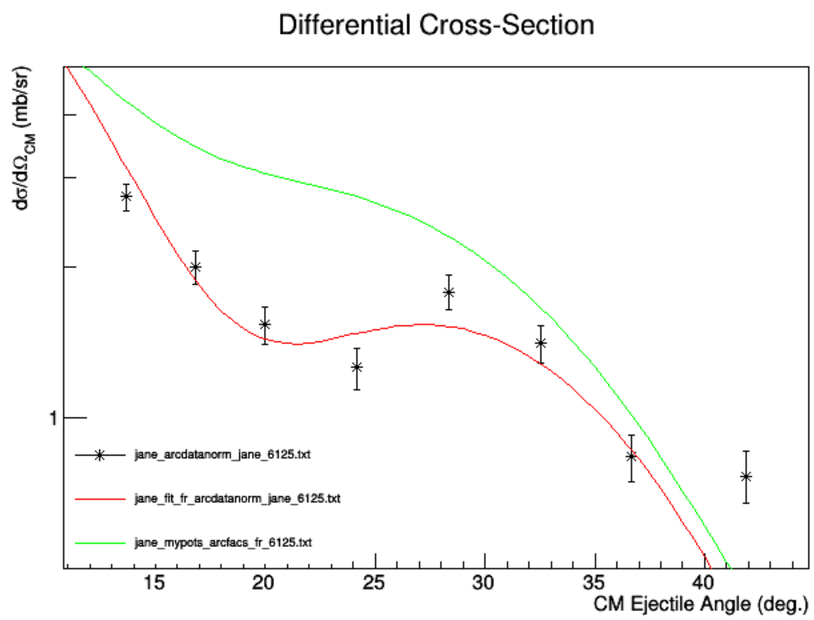

FIG. 6. ^{25}Mg$(d,p)^{25}$Mg(6.125 MeV, 3^+) Comparison of fitted spectroscopic factors for the current work (red) and those of Arc. et al (green). Both use the finite-range approximation and the potentials of Tab. III.

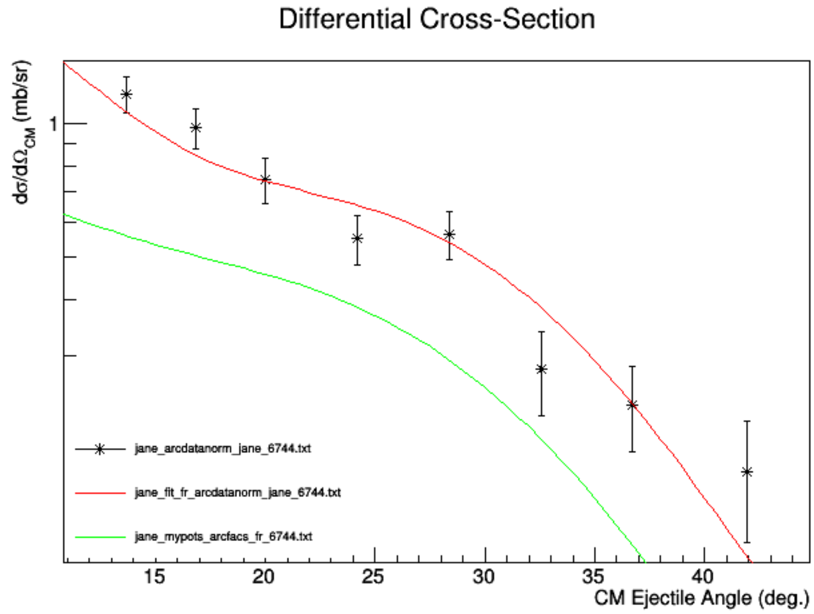

FIG. 7. ^{25}Mg$(d,p)^{25}$Mg(6.744 MeV, 2^+) Comparison of fitted spectroscopic factors for the current work (red) and those of Arc. et al (green). Both use the finite-range approximation and the potentials of Tab. III.

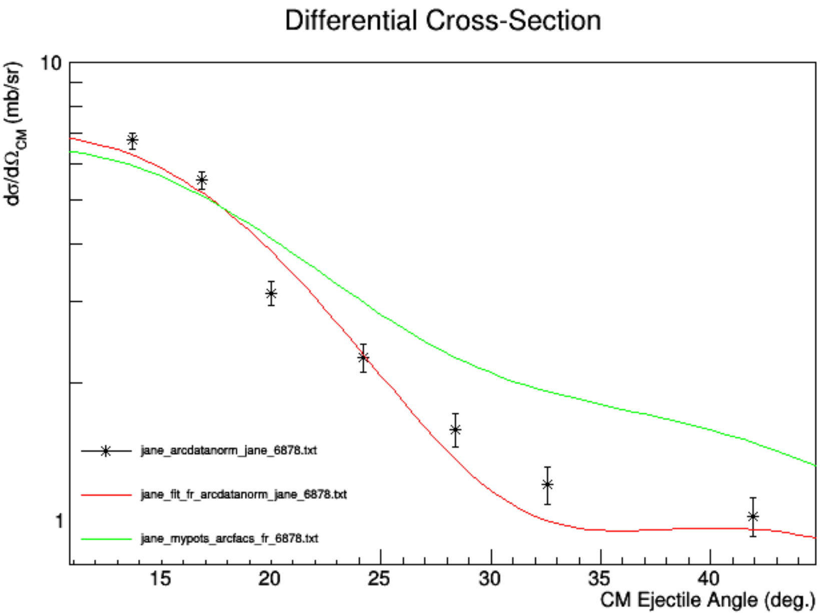

FIG. 8. ^{25}Mg$(d,p)^{25}$Mg(6.878 MeV, 3^-) Comparison of fitted spectroscopic factors for the current work (red) and those of Arc. et al (green). Both use the finite-range approximation and the potentials of Tab. III.

III. DEUTERON BREAKUP EFFECTS

The effects of deuteron breakup were investigated through the Continuum-Discretized Coupled Channels (CDCC) method [8]. This effect has been shown to have significant effects on scattering and transfers involving the deuteron, and must be checked.

The $p - n$ relative motion was truncated at $l_{max} = 2$. Higher orders of angular momentum have been shown to be genereally insignificant. The potentials used are the same as detailed above for the FR-DWBA calculation, but that in place of the $d +^{25}$ Mg potential a potential each for $n/p +^{25}$ Mg has been substituted at half the incident deuteron energy. These single-nucleon potentials have been taken from [9].

For the relative *linear* $p - n$ momentum (which is continuous), a cutoff must also be chosen. Yahiro et al. [8] showed that a value of k_{max} =∼$0.5 fm^{-1}$ is sufficient for most cases, with a continuum bin width of Δk =$0.125 fm^{-1}$. To probe the effects of the cutoff on the current reaction, values of k_{max} =0.250, 0.375, and $0.500 fm^{-1}$ were investigated (corresponding to 2, 3, and 4 bins respectively). Comparison to the FR-DWBA results for the elastic scattering and transfer to the ground state, 1.809 MeV, 6.125 MeV, 6.744 MeV, and 6.878 MeV states are shown in Fig's 9–14.

The modeling of deuteron breakup has a notable effect on some cases (e.g. transfer to the ^{25}Mg ground state), but not on others. In particular the states of interest for the current work are generally minimally affected by the inclusion of the breakup channels. Any differences in cross section (and therefore extracted spectroscopic factors) are on the order of the data uncertainty in the measured angular region. Therefore, for the time being (i.e. investigation of multi-step transfer components via coupled channels), the deuteron breakup effect will be neglected.

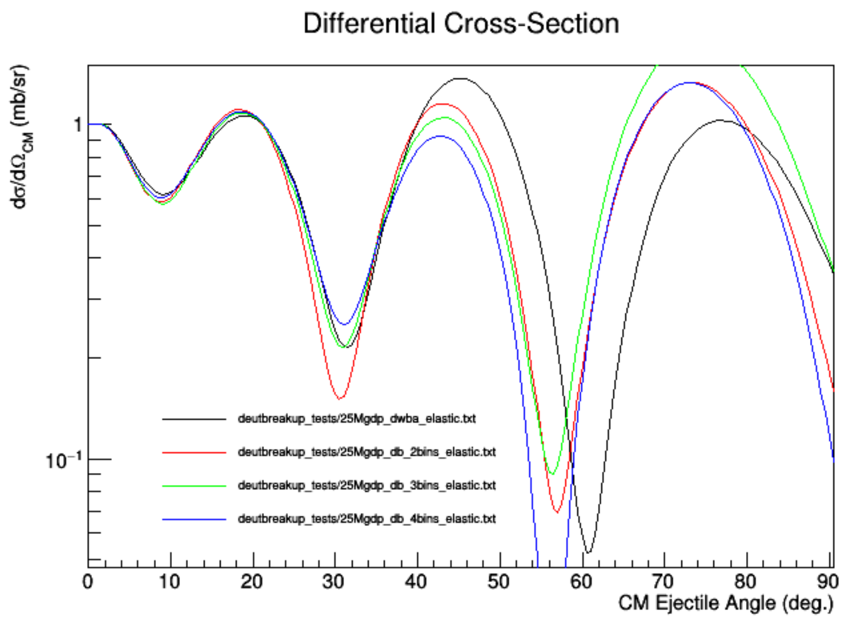

FIG. 9. ^{25}Mg(d,d) (elastic Ratio-to-Rutherford) Comparison of finite-range DWBA (black) to the treatment of the deuteron breakup channels via CCBA at various $p-n$ linear momentum cutoffs (in fm^{-1}): 0.250 (red), 0.375 (green), and 0.500 (blue). In general, a higher cutoff than those shown here would be preferred such that the sensitivity of the scattering to the CDCC is minimal. Fitted FR-DWBA amplitudes are used.

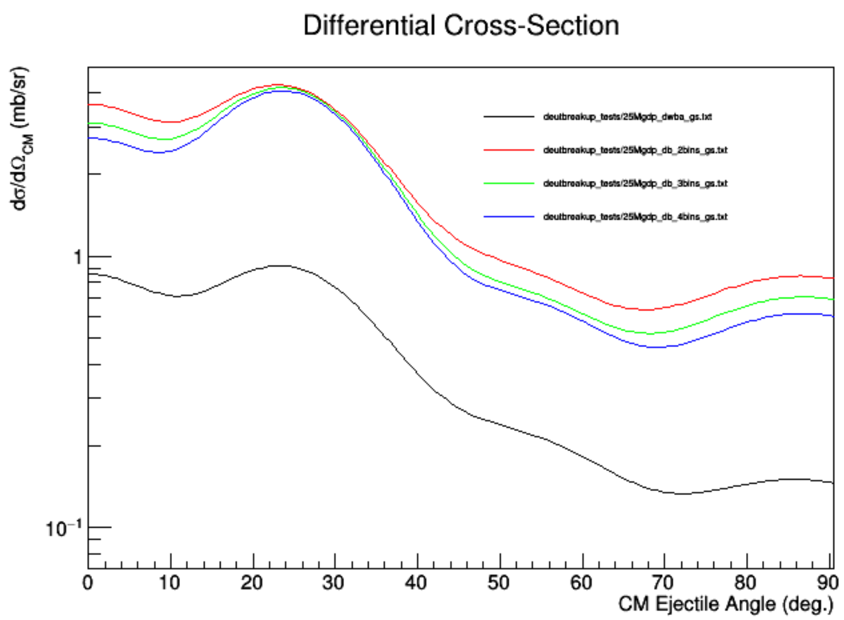

FIG. 10. ^{25}Mg$(d,p)^{25}$Mg(g.s., 0^+) Comparison of finite-range DWBA (black) to the treatment of the deuteron breakup channels via CCBA at various $p-n$ linear momentum cutoffs (in fm^{-1}): 0.250 (red), 0.375 (green), and 0.500 (blue). Fitted FR-DWBA amplitudes are used.

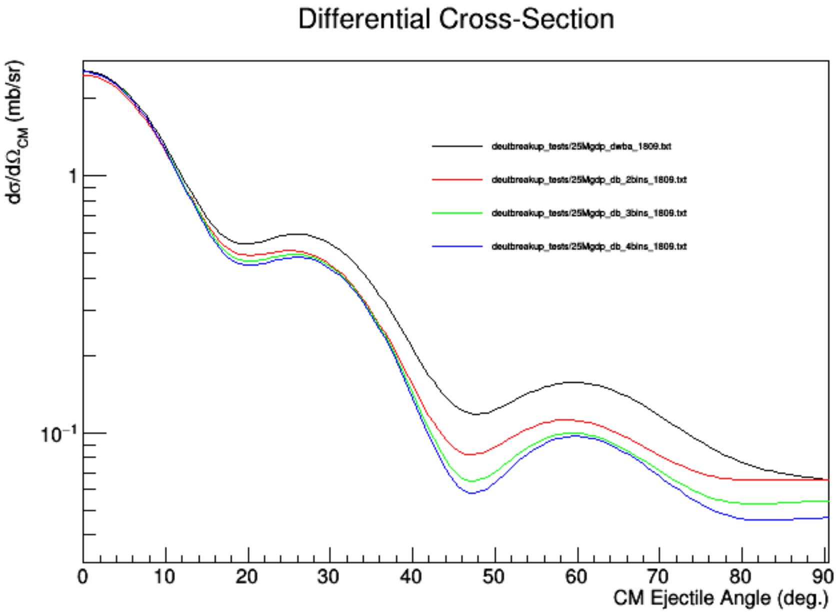

FIG. 11. ^{25}Mg$(d,p)^{25}$Mg(1.809 MeV, 2^+) Comparison of finite-range DWBA (black) to the treatment of the deuteron breakup channels via CCBA at various $p-n$ linear momentum cutoffs (in fm^{-1}): 0.250 (red), 0.375 (green), and 0.500 (blue). Fitted FR-DWBA amplitudes are used.

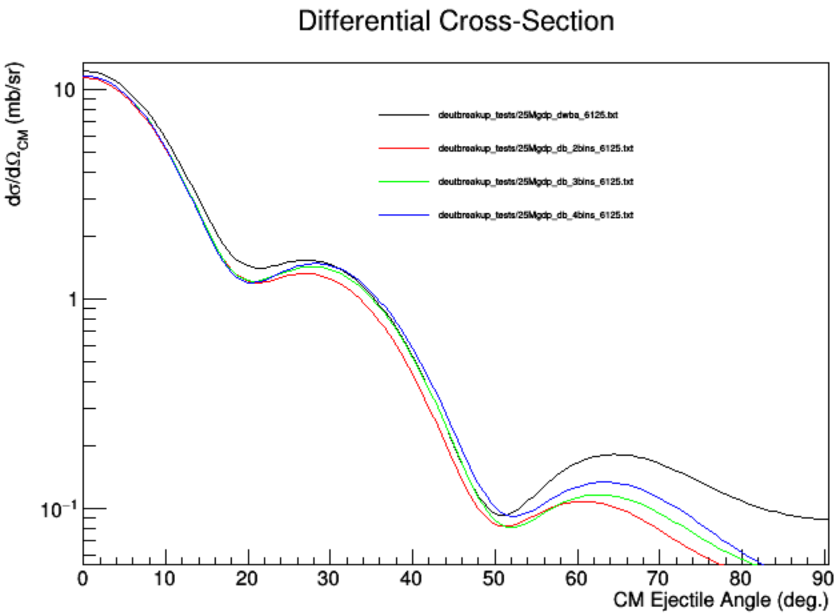

FIG. 12. ^{25}Mg$(d,p)^{25}$Mg(6.125 MeV, 3^+) Comparison of finite-range DWBA (black) to the treatment of the deuteron breakup channels via CCBA at various $p-n$ linear momentum cutoffs (in fm^{-1}): 0.250 (red), 0.375 (green), and 0.500 (blue). Fitted FR-DWBA amplitudes are used.

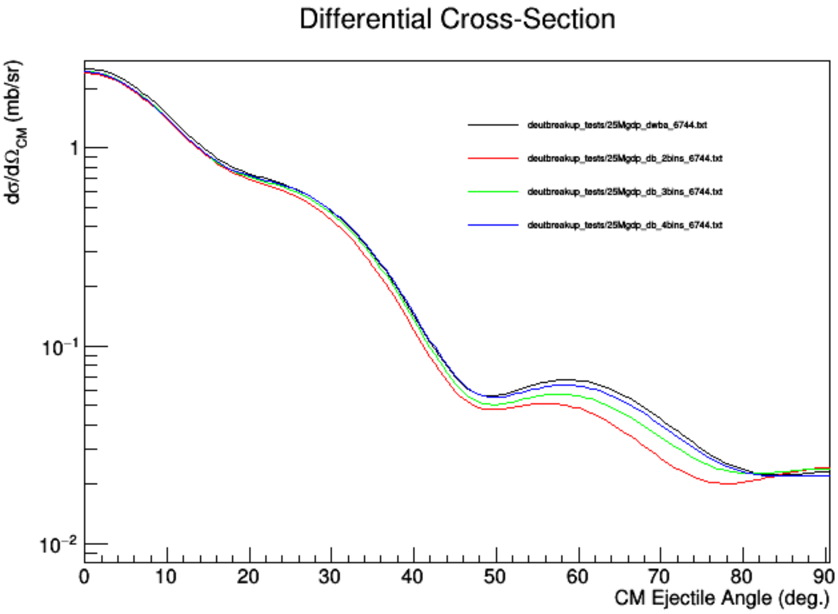

FIG. 13. ^{25}Mg$(d,p)^{25}$Mg(6.744 MeV, 2^+) Comparison of finite-range DWBA (black) to the treatment of the deuteron breakup channels via CCBA at various $p-n$ linear momentum cutoffs (in fm^{-1}): 0.250 (red), 0.375 (green), and 0.500 (blue). Fitted FR-DWBA amplitudes are used.

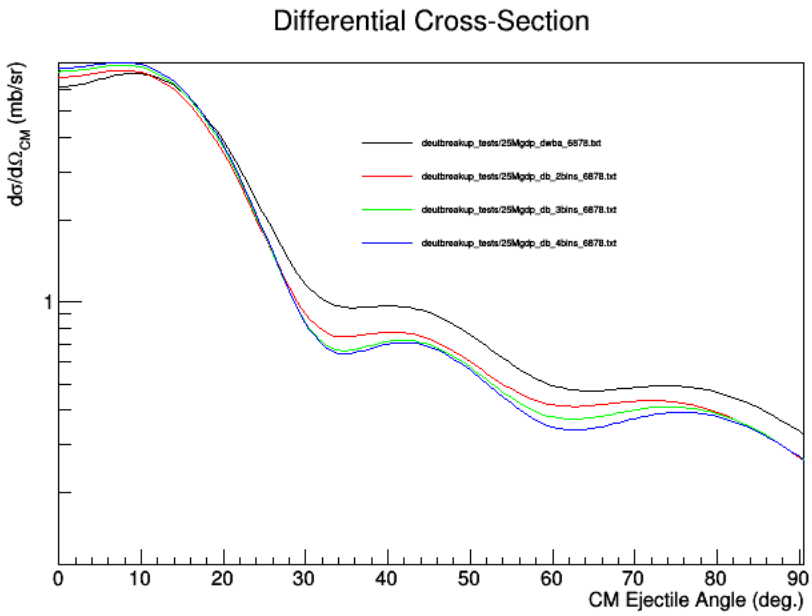

FIG. 14. ^{25}Mg$(d,p)^{25}$Mg(6.878 MeV, 3^-) Comparison of finite-range DWBA (black) to the treatment of the deuteron breakup channels via CCBA at various $p-n$ linear momentum cutoffs (in fm^{-1}): 0.250 (red), 0.375 (green), and 0.500 (blue). Fitted FR-DWBA amplitudes are used.

IV. MULTI-STEP (COUPLED CHANNELS) TREATMENT

To motivate the selection of relevant multi-step channels via inelastic excitation of the ^{25}Mg target, shell model calculations using `CoSMo` [5] in the new FSU interaction were performed. Only positive parity states in the target have been included. Those spectroscopic overlaps based on the ^{25}Mg ground state are included in Tab. II for comparison to the data and previous calculations. Since no spin information was available from the `CoSMo` results, the only differentiation between components is the orbital l.

For the purposes of selecting out strong potential multi-step components of the states of interest, those components calculated by `CoSMo` to have spectroscopic strength greater than 0.05 are shown in Tab. IV. Also given are the signed amplitudes corresponding to these factors.

Note that several of the amplitudes have a relative negative sign (phase) compared to others in their wave function. The effect of this sign change is negligible for the single-step amplitudes already fit. Nevertheless, going forward into the multi-step regime the relative signs as suggested by `CoSMo` will be retained.

The Coupled-Channels Born Approximation (CCBA) will be used for the time being. This treats couplings of the different target excitations in full CC, then the transfer itself in the BA (i.e. no back coupling).

To describe the inelastic excitations to these various states, the rotational model native to `FRESCO` will be used. These are specified by couplings of the ground state to various excited state deformed potentials dependent on the $B(Ek)$ values of the electromagnetic transitions. The $5/2^+$ nature of the ^{25}Mg ground state ensures that quadrupole $B(E2)$ values suffice for each excited state of interest. To start, these transition matrix elements will be taken from the NNDC database [10]. They are tabulated in Tab. V and include conversions into the reduced form $M(E2)$ used by `FRESCO` for explicit model-independent couplings. The nuclear deformation lengths $\delta = \beta R_{def}$ as they would be related to the Coulomb $M(E2)$ in the rotational model are included as well.

As part of the inelastic coupling mechanism, the band structure of the target must be considered. The list of states in Tab. IV correspond to two rotational bands in ^{25}Mg identified by the NNDC (15). These bands are rooted on 1) the $5/2^+$ ground state ($K = 5/2$) which includes the $7/2^+(1)$ and $9/2^+(1)$ and 2) the $1/2^+$ first excited state ($K = 1/2$) which includes the $3/2^+(1)$, $5/2^+(2)$, $7/2^+(2)$, and $9/2^+(2)$. No bands are identified with the $1/2^+(2)$ and $3/2^+(2)$.

Despite the separate bands, all states of interest for inelastic excitation have direct decay paths to the ground state, as can be seen by the nonzero transition probabilities in Tab. V.

It is also important to keep in mind what N. Keeley et al. [11] call "kinematic" effects on wave function measurements in direct reactions. These are such things as the reaction Q-value, possible momentum transfers, and inelastic cross sections in general all at the reaction energy. Consider for example the measurement by Emmerich et al. [12] of inelastic deuteron scattering off of ^{25}Mg at 10.1 MeV. The $7/2^+(1)$ and $9/2^+(1)$ are populated with much larger ($\sim$ order of magnitude) cross sections than the rest. This is consistent with the ground state band picture described above. The `CoSMo` FSU interaction predicts strong overlap of these states with the first excited state, but not those higher lying.

Because of the temptingly simple band structure two-step picture for the first excited state, transfer to this state will first be examined using the rotational model. Under this model the reduced matrix elements $M(E2)$ become *intrinsic* reduced elements $Mn(E2)$ which has a common value of $\sim$17(2) e^2fm^4 for both direct excitations. This corresponds to a joint nuclear deformation length of $\delta = 1.6(2)$ fm. Unless otherwise specified these quantities will be positive, i.e. assuming a prolate deformed nature. All possible quadrupole couplings within the specified band are included in the model. Vibrational modes and reconfigurations are neglected.

Fig. 16 displays (in green) such a CCBA calculation using the strict, signed `CoSMo` amplitudes. Also shown are the original DWBA fit (blue) and the fit of this same CCBA rotational band hypothesis (red) described later.

One of the questions to answer with this analysis is whether such multi-step excitations are discernible from one another and hence fitable. Fig. 17 directly compares transfer steps to the 1.809 MeV 2^+ from each of the first three states in the ground state rotational band. Each step has been arbitrarily given a unity core-composite overlap. In accordance with the shell model wavefunction predictions, each step assumes a $0d_{5/2}$ transfer nucleon configuration.

Due to A) the similar qualitative shape of each d-wave and B) the relative featurelessness of the multi-step components, it will be difficult to "fit" reliably spectroscopic amplitudes given the current data set.

Even so, a fit using coupled channels in the ground state rotational band was attempted for the 1.809 MeV 2^+. The $1s_{1/2}$ component built off of the target ground state was included in the fit, while all the rest were $0d_{5/2}$ as motivated by `CoSMo`. The result (compared to `CoSMo` predictions) are given in Tab. VI and in Fig. 16 as the red curve. Relative signs were fixed in accordance with `CoSMo` predictions.

The results displayed in Tab. VI demonstrate that such a CCBA fit is unsuccessful. It cannot differentiate the $0d_{5/2}$

natures of the inelastic channels in a way that makes them statistically different from zero. Note that this is feature is not restricted to the 1.809 MeV 2^+: all ^{26}Mg states of interest have competing s- and d-wave multi-step components which will be as indistinguishable in finite-range CCBA or CRC.

The chief obstacle in this is that the measured angular range is already qualitatively well described by the single-step DWBA (compare the red CCBA and blue DWBA fits in Fig. 16). This suggests negligible contribution from odd l (p,f,...) nucleon transfers from negative parity ^{25}Mg states (the first one of which is the $3/2^-$ at 3.413 MeV, hence higher in energy and less likely than those states already considered). The quantitative discrepancies in the s- and d-wave spectroscopic factors, meanwhile, prove difficult to rectify in the coupled channels model due to the inability to distinguish even strong coupling channels from one another in a fit to data. The lack of elastic and inelastic data for the current experiment also makes the validity of the included coupled channels difficult to verify.

^{26}Mg	^{25}Mg	nL_j^ν	Ampl.	C^2S
1.897 MeV $2^+(1)$	0.000 MeV $5/2^+(1)$	$0d_{5/2}$	-0.5898	0.3478
	1.098 MeV $3/2^+(1)$	$1s_{1/2}$	-0.2567	0.0659
	1.720 MeV $7/2^+(1)$	$0d_{5/2}$	1.0443	1.0906
	1.996 MeV $5/2^+(2)$	$1s_{1/2}$	-0.3083	0.0951
		$0d_{5/2}$	0.2753	0.0758
	2.812 MeV $3/2^+(2)$	$0d_{5/2}$	0.2579	0.0665
	3.447 MeV $9/2^+(1)$	$0d_{5/2}$	-0.7544	0.5692
6.180 MeV $3^+(3)$	0.000 MeV $5/2^+(1)$	$1s_{1/2}$	0.3757	0.1412
		$0d_{3/2}$	-0.5294	0.2803
	0.606 MeV $1/2^+(1)$	$0d_{5/2}$	0.2622	0.0688
	1.996 MeV $5/2^+(2)$	$0d_{5/2}$	-0.3689	0.1361
	2.812 MeV $3/2^+(2)$	$0d_{5/2}$	0.4978	0.2478
	2.902 MeV $7/2^+(2)$	$0d_{5/2}$	-0.3879	0.1505
	3.447 MeV $9/2^+(1)$	$0d_{3/2}$	0.2813	0.0791
	3.897 MeV $9/2^+(2)$	$0d_{5/2}$	-0.2554	0.0652
6.678 MeV $2^+(6)$	0.000 MeV $5/2^+(1)$	$0d_{5/2}$	0.3155	0.0995
	0.606 MeV $1/2^+(1)$	$0d_{3/2}$	0.4480	0.2007
	1.996 MeV $5/2^+(2)$	$1s_{1/2}$	-0.4594	0.2111
		$0d_{3/2}$	0.3883	0.1507
	2.584 MeV $1/2^+(2)$	$0d_{3/2}$	0.2308	0.0533
		$0d_{5/2}$	0.2835	0.0804
	2.812 MeV $3/2^+(2)$	$1s_{1/2}$	-0.6162	0.3797
		$0d_{5/2}$	0.2762	0.0763
6.786 MeV $3^-(1)$	0.000 MeV $5/2^+(1)$	$1p_{3/2}$	-0.2843	0.0808
		$0f_{7/2}$	0.5926	0.3512
	0.606 MeV $1/2^+(1)$	$0f_{7/2}$	-0.2696	0.0727

TABLE IV. (^{26}Mg|^{25}Mg) single-particle overlaps calculated by `CoSMo` [5] in the full $sd-fp$ shell FSU interaction. Only those with spectroscopic strength greater than 0.05 are shown. The excitation energies are those from `CoSMo`.

^{25}Mg(E_{exc}, I)	$B(E2; I \to g.s.)$ e^2fm^4	$M(E2)$ $e$fm^2	δ (fm)
(585 keV, $1/2^+(1)$)	2.445(43)	2.211(19)	0.2030(17)
(975 keV, $3/2^+(1)$)	3.95(57)	3.97(29)	0.365(27)
(1612 keV, $7/2^+(1)$)	104(26)	28.8(36)	2.64(33)
(1965 keV, $5/2^+(2)$)	1.95(52)	3.42(46)	0.314(42)
(2563 keV, $1/2^+(2)$)	17(13)	5.8(22)	0.53(20)
(2738 keV, $7/2^+(2)$)	0.43(22)	1.9(5)	0.17(5)
(2801 keV, $3/2^+(2)$)	7.4(17)	5.4(6)	0.50(6)
(3405 keV, $9/2^+(1)$)	29.5(39)	17.2(11)	1.58(10)

TABLE V. Tabulated values from the NNDC [10] of quadrupole transition matrix elements $B(E2)$ between the $5/2^+$ ground state and relevant excited states of the ^{25}Mg target. As given in the database, these are for decay down to the ground state. Also included are the reduced form $M(E2)$ and rotational model deformation length $\delta = \beta R_{def}$ used as inputs to `FRESCO`. Both of these quantities can be (jointly) positive (prolate) or negative (oblate) and are symmetric between decay to and excitation from an initial state.

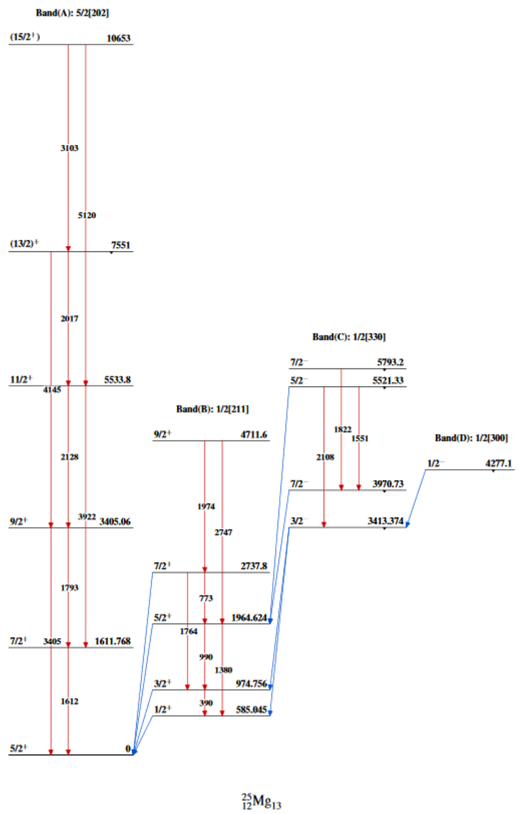

FIG. 15. Rotational band structure of the target ^{25}Mg pulled from the NNDC [10].

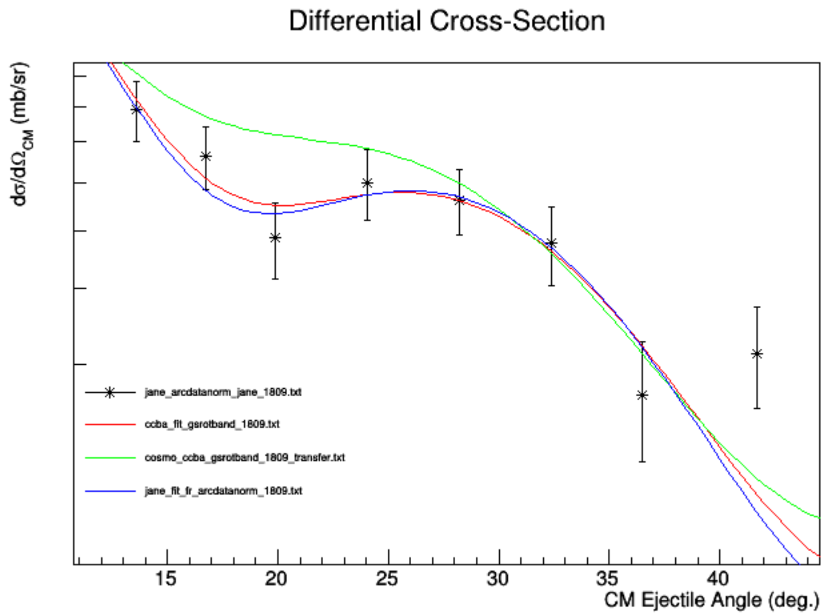

FIG. 16. ^{25}Mg$(d,p)^{25}$Mg(1.809 MeV, 2^+) Multi-Step CCBA transfer using (^{26}Mg$|^{25}$Mg) overlaps from: the CCBA fit (red), the DWBA fit (blue), and `CoSMo` (green). Only the first three states of the ground state rotational band (ground state inclusive) are considered. All quadrupole couplings between these states are allowed within the rotational model. The entrance and exit channel potentials are still those of the *un*coupled optical model.

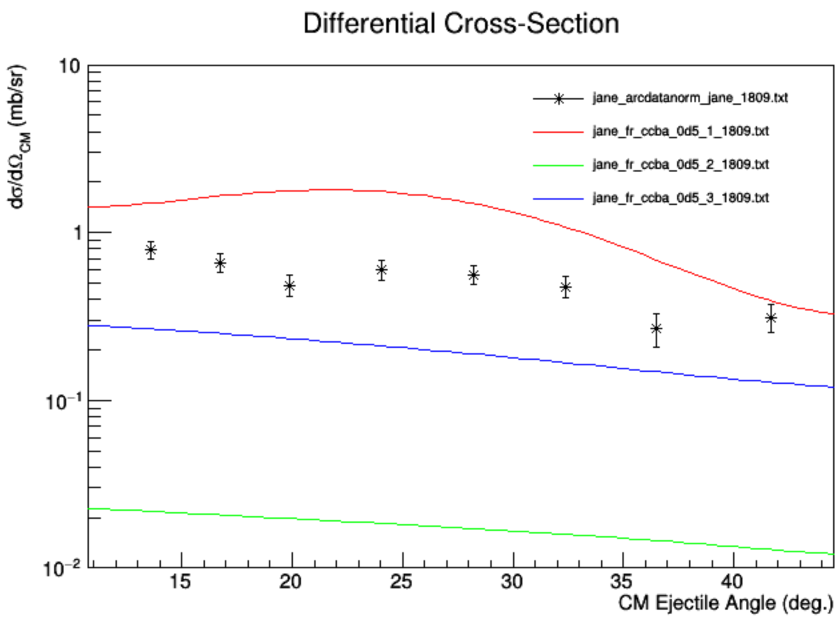

FIG. 17. ^{25}Mg$(d,p)^{25}$Mg(1.809 MeV, 2^+) Direct comparison of unity-amplitude cross-section contributions of each inelastic excitation in the ground state rotational band (up to the first $9/2^+$). Each assumes a $0d_{5/2}$ nucleon transfer in accordance with the shell model wavefunction predictions. Red: $5/2^+(1)$ (ground state); green: $7/2^+(1)$; blue: $9/2^+(1)$

1.806 MeV (2^+)	CCBA Fit		CoSMo		DWBA Fit	
	Ampl.	C^2S	Ampl.	C^2S	Ampl.	C^2S
$5/2^+ \otimes 1s_{1/2}$	0.2521(427)	0.0636(215)	0.1847	0.0341	0.2659(370)	0.0707(197)
$5/2^+ \otimes 0d_{5/2}$	-0.5040(390)	0.2540(393)	-0.5898	0.3478	0.4840(387)	0.2343(375)
$7/2^+ \otimes 0d_{5/2}$	0.0000(7200)	0.0000(–)	1.0443	1.0906	–	–
$9/2^+ \otimes 0d_{5/2}$	-0.4306(2849)	0.1854(–)	-0.7544	0.5692	–	–
	Fit $\chi^2/N = 0.89$				Fit $\chi^2/N = 0.99$	

TABLE VI. Fitted spectroscopic (^{25}Mg/^{26}Mg) overlaps for the 1.809 MeV 2^+ in a CCBA hypothesis including the first three states (ground state inclusive) of the ground state rotational band in the target ^{25}Mg. A rotational model was used with reduced matrix elements and deformation lengths calculated from the NNDC database $B(E2)$ values. Comparison to the contemporaneous shell model results via `CoSMo` and the original DWBA fit using only the single-step components. Relative signs for the CCBA fit were fixed in accordance with `CoSMo` predictions. The uncertainties indicated by (–) are too large for first order error propagation.

V. UNCERTAINTY VISUALIZATION IN FR-DWBA SPECTROSCOPIC FACTORS

If we take that the FR-DWBA results presented above are the most reliable of the sets considered, it becomes relevant to inspect the effect of the uncertainties on the spectroscopic factors tabulated in Tab. II. These uncertainties were determined from the fits themselves, namely from `FRESCO`'s use of the `MINUIT` fitting routines. Below are shown the FR-DWBA angular distributions for the states of interest holding one component fixed while varying the other to the extremes of its tabulated uncertainty.

Also included are extra large bounds given a rough overall systematic uncertainty of 15% (on *top* of the uncertainty from the fit). The various specfac values used are shown in Tab. VII.

Variation of s/p-wave components (where relevant) holding d/f components constant are shown in Fig's 18–21. The opposite case (varying d/f holding s/p constant) are in Fig's 22–25.

	C^2S	Amplitude
1.806 MeV (2^+)		
$1s_{1/2}$	0.0707(197)(303)	$0.2659^{+0.0348,+0.0519}_{-0.0401,-0.0649}$
$0d_{5/2}$	0.2343(375)(726)	$-0.4840^{+0.0404,+0.0819}_{-0.0373,-0.0700}$
6.125 MeV (3^+)		
$1s_{1/2}$	0.1743(170)(431)	$0.4175^{+0.0199,+0.0488}_{-0.0209,-0.0553}$
$0d_{3/2}$	0.2336(235)(585)	$-0.4833^{+0.0249,+0.0649}_{-0.0238,-0.0572}$
6.744 MeV (2^+)		
$1s_{1/2}$	0.0397(136)(196)	$-0.1992^{+0.0376,+0.0574}_{-0.0387,-0.0443}$
$0d_{5/2}$	0.1612(165)(407)	$0.4015^{+0.0200,+0.0478}_{-0.0211,-0.0544}$
6.878 MeV (3^-)		
$1p_{3/2}$	0.2338(95)(446)	$-0.4835^{+0.0099,+0.0485}_{-0.0098,-0.0441}$
$0f_{7/2}$	0.0642(319)(415)	$0.2534^{+0.0566,+0.0717}_{-0.0737,-0.1027}$

TABLE VII. Variation of statistical and systematic uncertainties for the FR-DWBA fits of spectroscopic factors. In the first parentheses are the uncertanties purely from the fit; in the second is included a 15% overall systematic addition. For ease of reference the corresponding amplitudes are also included.

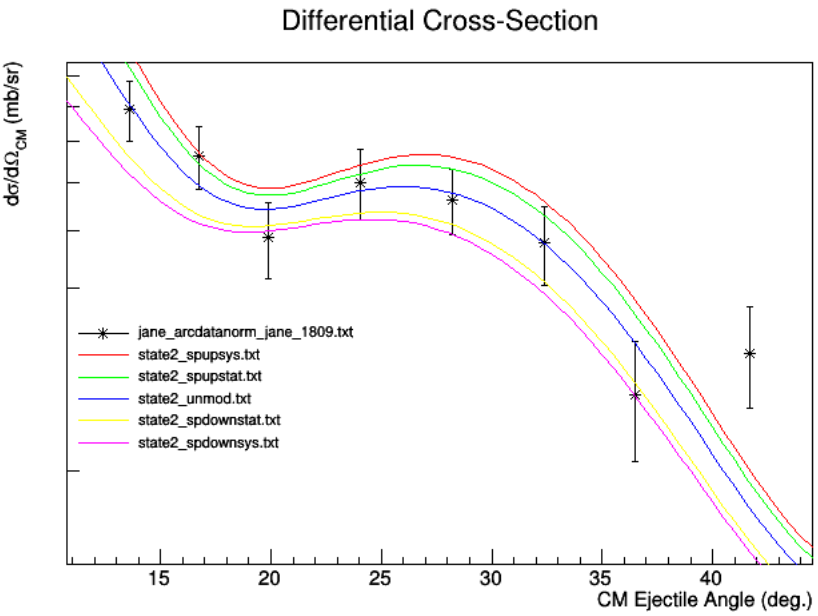

FIG. 18. ^{25}Mg(d,p)^{25}Mg(1.809 MeV, 2^+) s-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding d-wave component fixed to tabulated value.

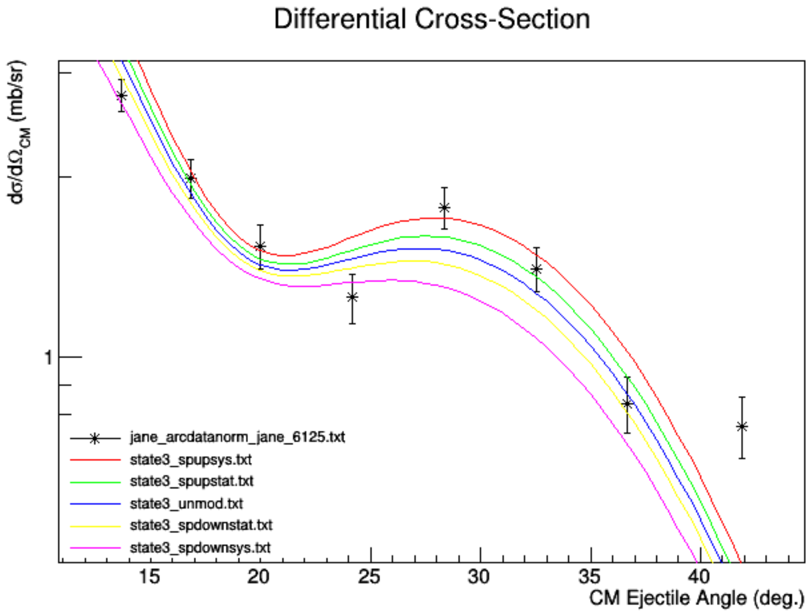

FIG. 19. ^{25}Mg$(d,p)^{25}$Mg(6.125 MeV, 3^+) s-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding d-wave component fixed to tabulated value.

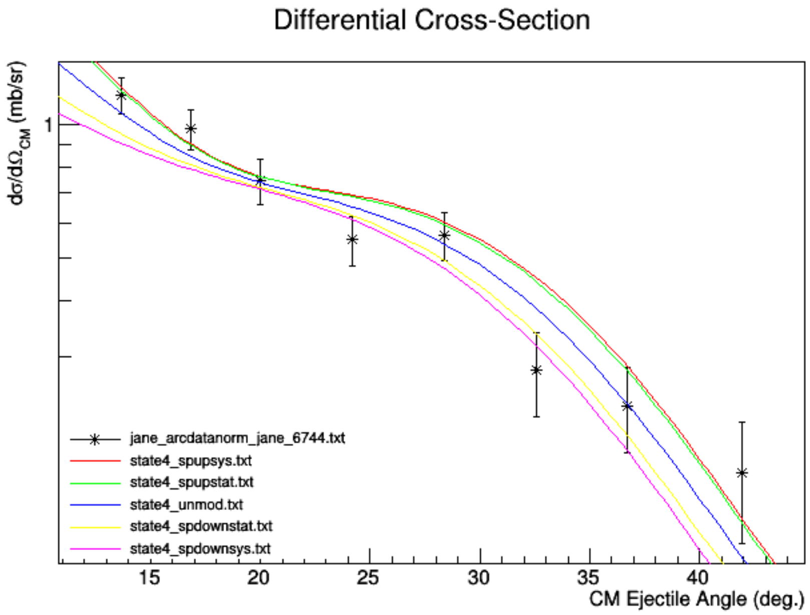

FIG. 20. ^{25}Mg$(d,p)^{25}$Mg(6.744 MeV, 2^+) s-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding d-wave component fixed to tabulated value.

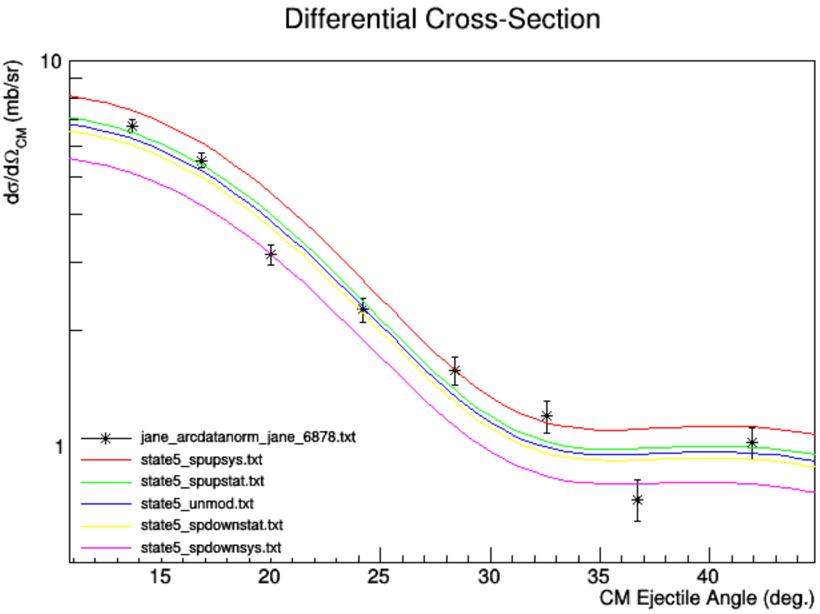

FIG. 21. ^{25}Mg$(d,p)^{25}$Mg(6.878 MeV, 3^-) p-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding f-wave component fixed to tabulated value.

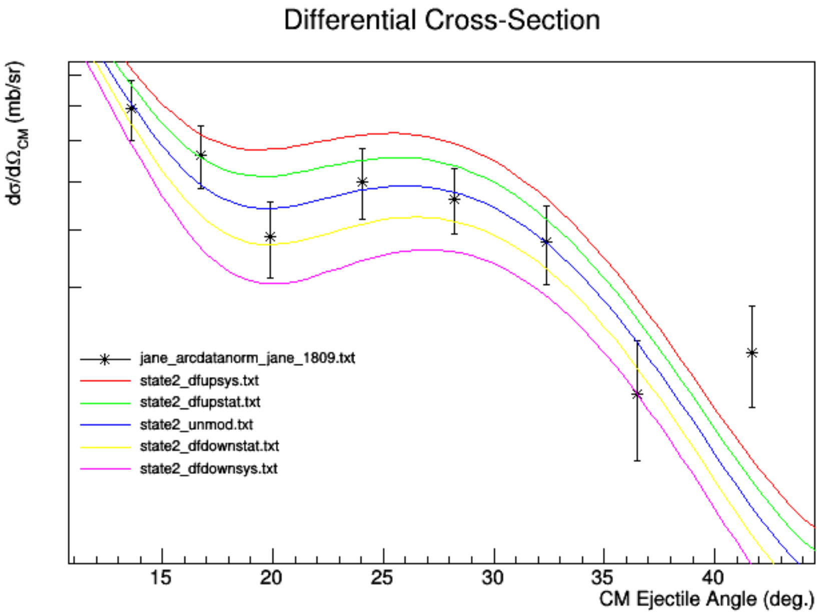

FIG. 22. ^{25}Mg$(d,p)^{25}$Mg(1.809 MeV, 2^+) d-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding d-wave component fixed to tabulated value.

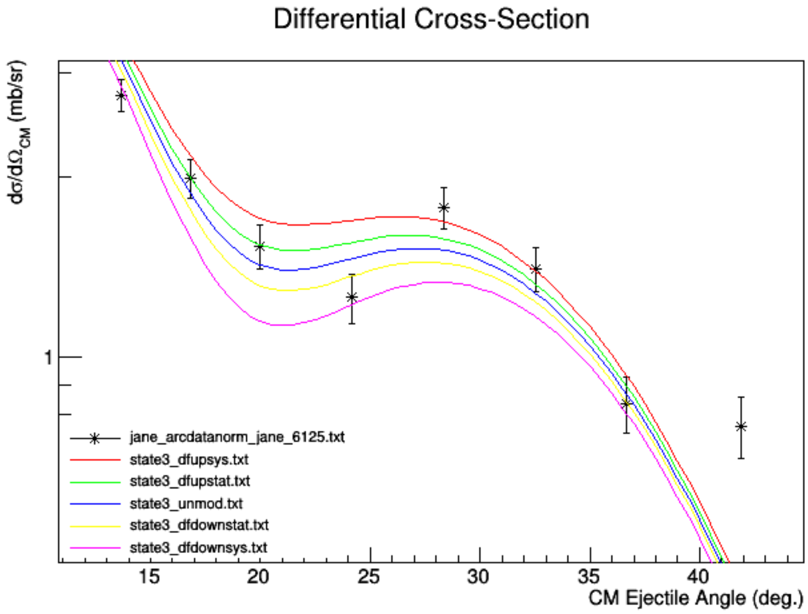

FIG. 23. ^{25}Mg$(d,p)^{25}$Mg(6.125 MeV, 3^{+}) d-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding d-wave component fixed to tabulated value.

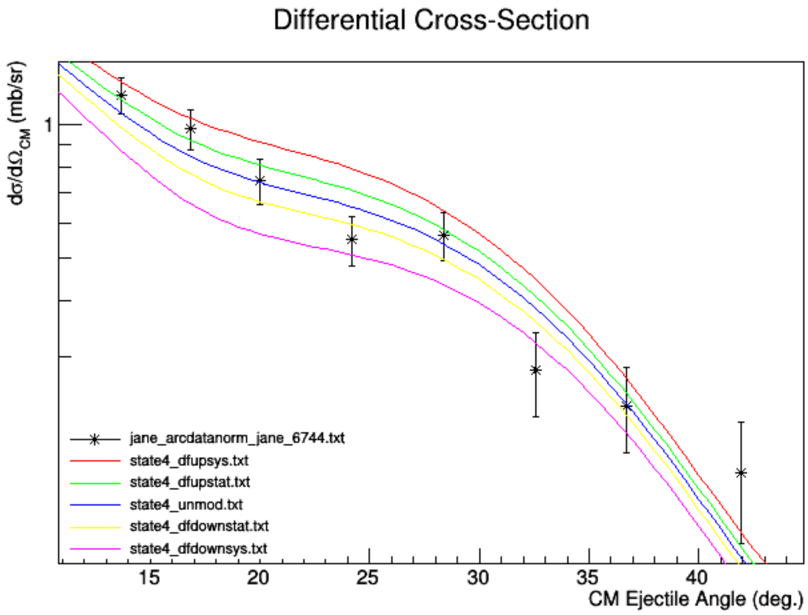

FIG. 24. ^{25}Mg$(d,p)^{25}$Mg(6.744 MeV, 2^{+}) d-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding d-wave component fixed to tabulated value.

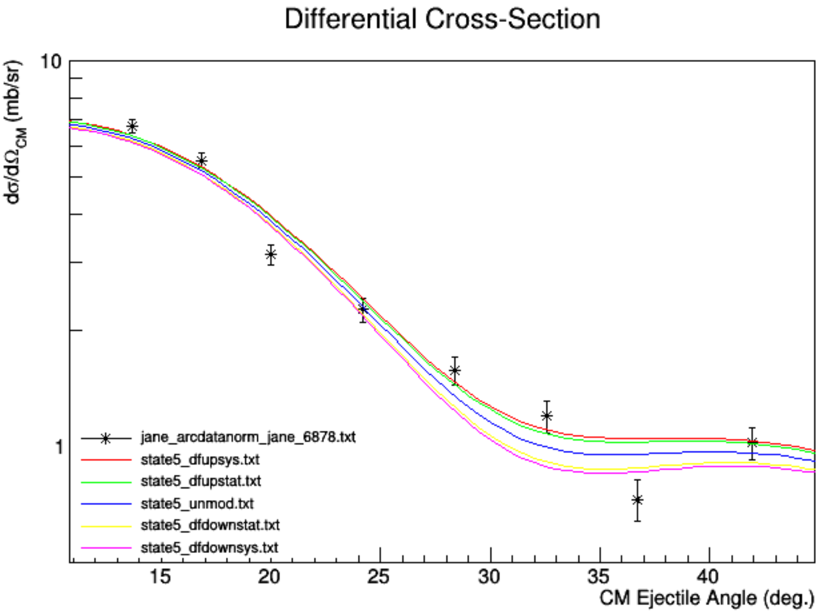

FIG. 25. ^{25}Mg$(d,p)^{25}$Mg(6.878 MeV, 3^-) f-wave component specfac uncertainty visualization including: statistical + systematic (red and magenta), just systematic (green and yellow), and the fitted value itself (blue). Holding p-wave component fixed to tabulated value.

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